



Anglo-Chinese Junior College

JC2 Biology Preliminary Examination

Higher 2



A Methodist Institution
(Founded 1886)

CANDIDATE
NAME

FORM
CLASS

TUTORIAL
CLASS

INDEX
NUMBER

BIOLOGY

Paper 3 Long Structured and Free-response Questions

9744/03

30 August 2022
2 hours

Candidates answer on the Question Paper.
Additional Materials: Writing paper(s)

READ THESE INSTRUCTIONS FIRST

Write your Name, Class and Index number in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided in the question paper.

Section B

Answer any **one** question on the separate writing paper(s) provided.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all the writing paper(s) used securely together.

For Examiners' use only	
Section A	
1	/ 30
2	/ 10
3	/ 10
Section B	
4 or 5	/ 25
Total	/ 75

Section A

Answer **all** the questions in this section.

- 1** The Human Immunodeficiency Virus (HIV) is an enveloped virus that consists of glycoproteins on its surfaces. Fig. 1.1 shows the structure of an Env glycoprotein on HIV.

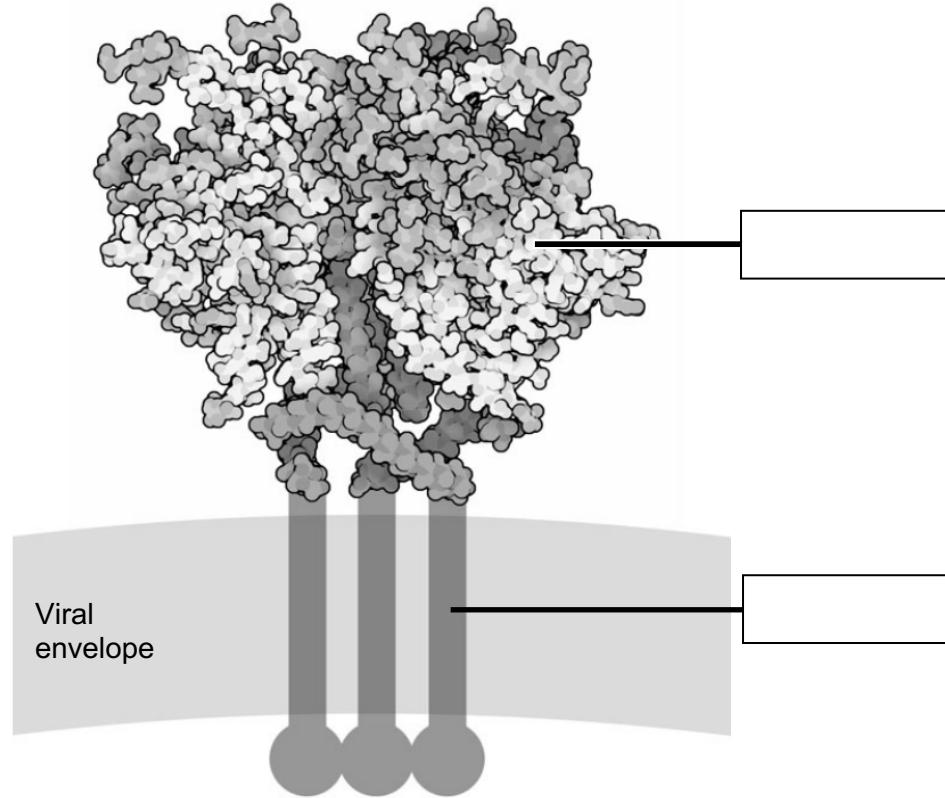


Fig. 1.1

- (a) (i)** The Env glycoprotein is made up of two associated subunits.

In the boxes in Fig. 1.1, state the name of these two subunits.

[1]

- (ii)** Describe how the glycoprotein interacts with the viral envelope.

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- (iii) HIV envelope proteins are glycosylated by host cell enzymes as HIV genome does not encode gene products capable of synthesising carbohydrates.

Suggest how glycosylation of HIV envelope proteins may impair the host immune response to HIV infection.

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- (b) Describe the interactions between HIV and the host cell to enable the entry of the virus into the cell.

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[Turn over

Upon entry into the host cell, HIV uses the enzyme reverse transcriptase to reverse transcribe its RNA genome into double-stranded DNA for integration into the host genome.

Reverse transcriptase is made up of two subunits, forming two different active sites that are essential for its function. Fig. 1.2 shows the location of these active sites on the enzyme with its newly formed double-stranded DNA.

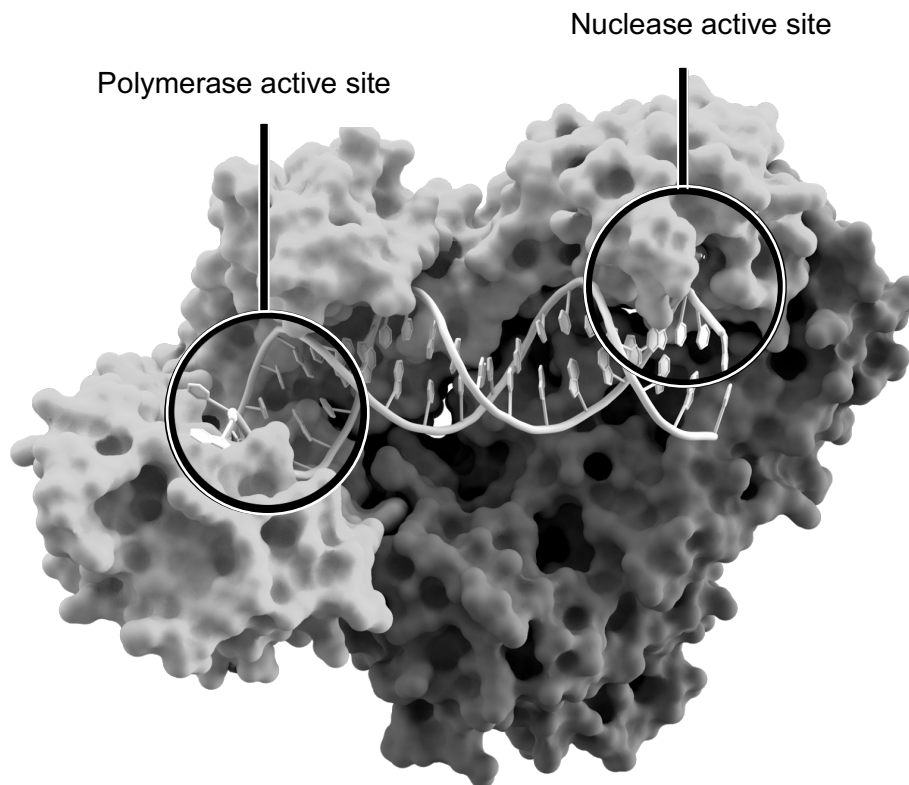


Fig. 1.2

- (c) (i) Explain what determines the precise positions of these two active sites in the structure of reverse transcriptase.

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- (ii) Compare the structures of reverse transcriptase and DNA.

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[Turn over

One application of reverse transcriptase in research is in a technique called reverse transcription-polymerase chain reaction (RT-PCR). The classical PCR technique can only be applied to DNA strands, whereas RT-PCR allows the application of PCR technique to an RNA template. A potential use of RT-PCR is in the study of gene expression.

Fig. 1.3 illustrates the process of RT-PCR using mRNA as a template. Oligo dT primers, which are single-stranded sequences of repeating deoxythymine, are added to the reaction mixture.

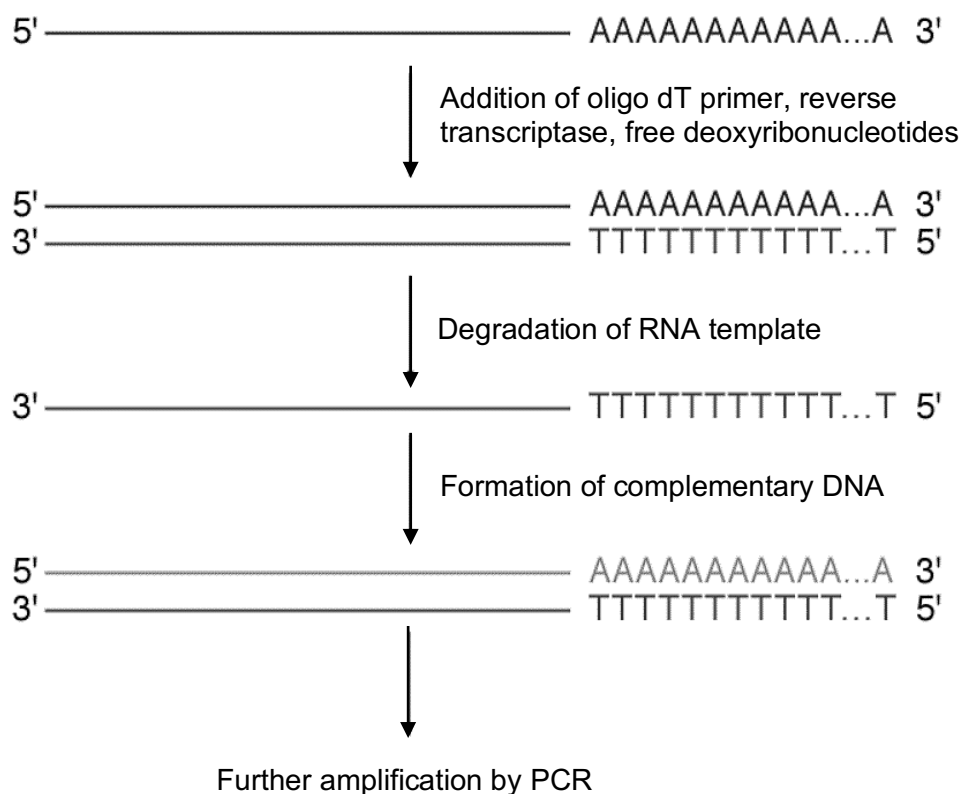


Fig. 1.3

- (d) (i) With reference to Fig. 1.2 and Fig. 1.3, explain how reverse transcriptase and oligo dT primers can be used in RT-PCR to study gene expression.

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The relative activity of reverse transcriptase at different temperatures can be investigated using the SYPRO orange fluorescence assay. The reverse transcriptase is incubated with a fluorescent chemical known as SYPRO orange and the temperature is gradually increased. SYPRO orange interacts with the hydrophobic regions in an enzyme which are normally not exposed. The amount of SYPRO orange which binds to an enzyme can be detected by measuring the intensity of the fluorescence.

Fig. 1.4 shows the results of the SYPRO orange fluorescence assay on two types of reverse transcriptase.

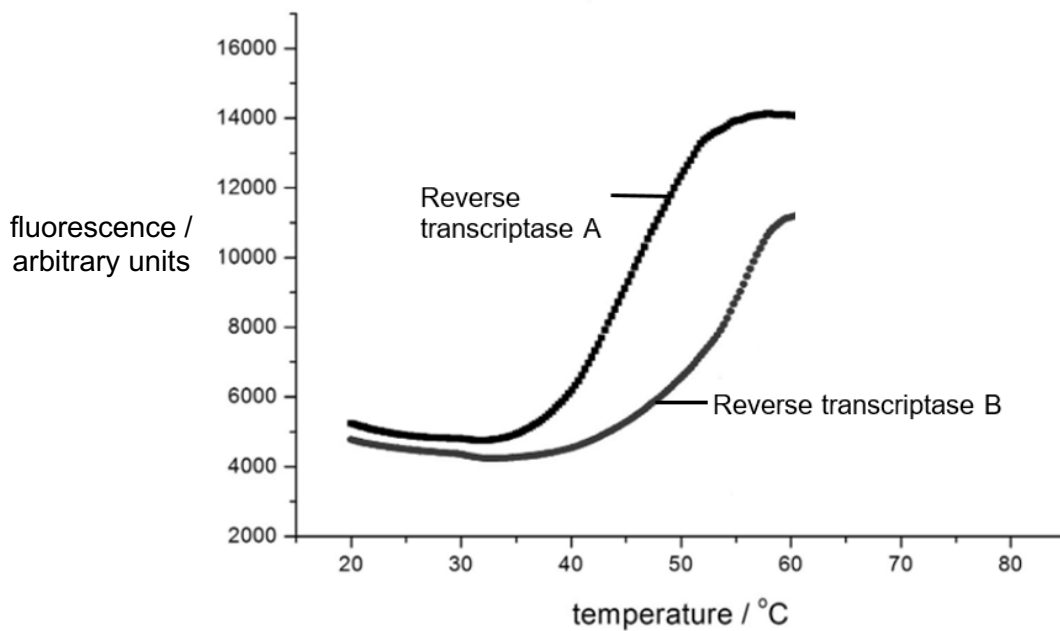


Fig. 1.4

- (ii) During RT-PCR, a high incubation temperature may be used.

Suggest and explain which reverse transcriptase is more suitable to be used for RT-PCR.

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[Turn over

Infections by some viruses are known to increase the risk of cancer development. The HIV weakens the immune system and increases the susceptibility of the body to infections which cause cancer. Other viruses like the human papillomavirus (HPV) can integrate its genome into the host genome and produce highly carcinogenic proteins.

Fig. 1.5 shows how a HPV infection which persists for many years can lead to cellular changes that, if untreated, may result in the formation of a malignant tumour.

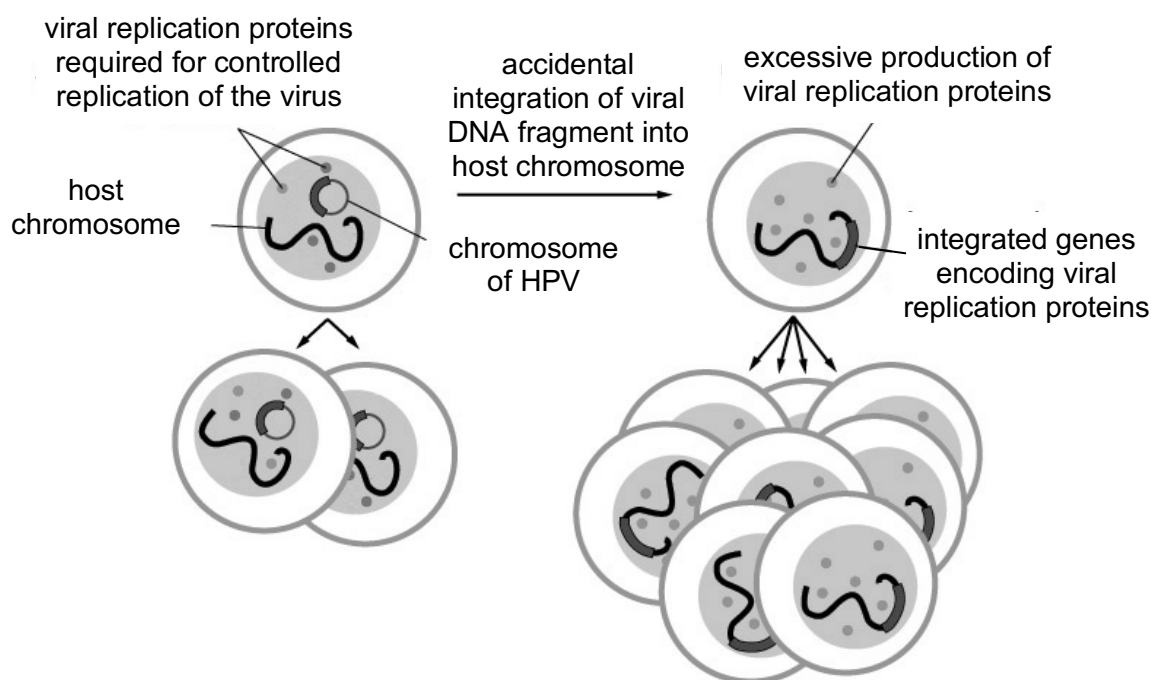


Fig. 1.5

Fig. 1.6(a) shows how proteins regulating the cell cycle checkpoints, such as p53 and Rb protein, keep the cell cycle in check and prevent cell proliferation in a normal cell. Fig. 1.6(b) shows an example of two viral replication proteins, E6 and E7, interfering with p53 and Rb protein function and resulting in excessive cell proliferation.

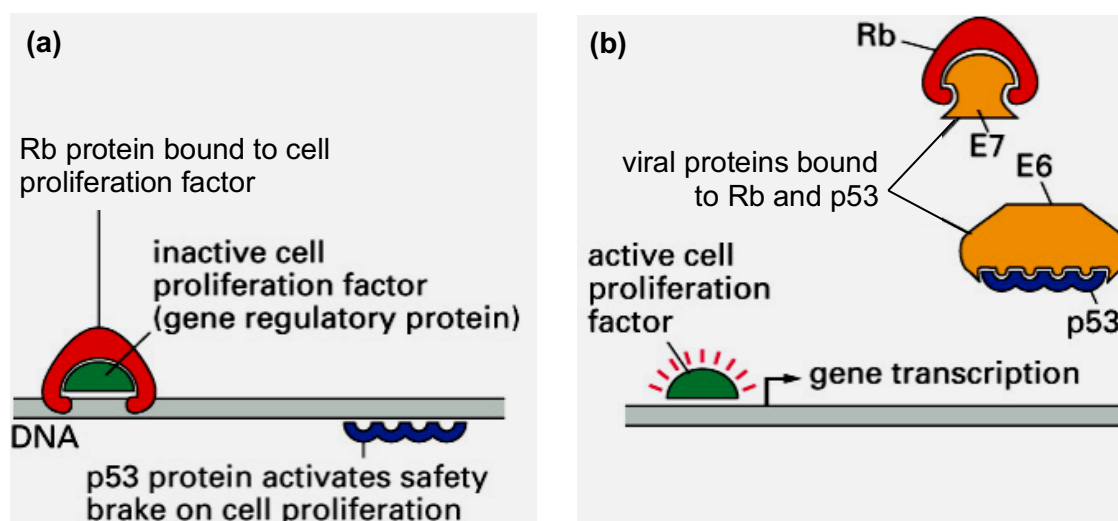


Fig. 1.6

- (e) (i) With reference to Fig. 1.5 and Fig 1.6, suggest and explain how a tumour can be formed in patients with HPV infection persisting for many years.

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- (ii) Suggest a reason why it may take 10 to 20 years for HPV-infected cells to become cancerous.

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HPVs spread easily through direct sexual contact, including vaginal, oral, and anal sex. Fig. 1.7 shows the HPV-associated cancer incidence in the United States between 2008 and 2012.

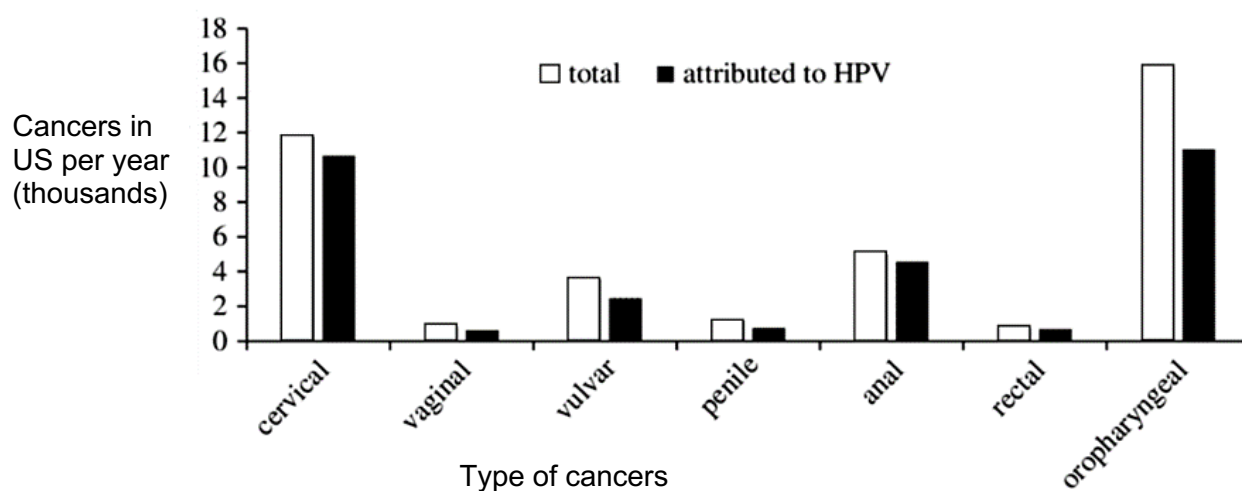


Fig. 1.7

[Turn over

The HPV vaccine was first introduced in 2006, with a recommendation to vaccinate females of ages 11 to 26. With growing evidence supporting the expansion of the vaccine, many countries are extending its coverage to males.

Fig. 1.8 shows the numbers of cancers caused by HPV and the gender distribution for each type of cancer in the United States each year.

Fig. 1.9 shows the estimated effect of HPV vaccination on oral HPV infections among individuals from 18 to 33 years of age in the United States population. Results are presented as the total number of infections in the absence of HPV vaccination, the number of preventable infections at 100% vaccination levels, and the number of potentially vaccine-prevented infections at current HPV vaccination levels in the population.

Fig. 1.10 shows the rates of HPV-associated cancers and age at diagnosis among men in the United States per year.

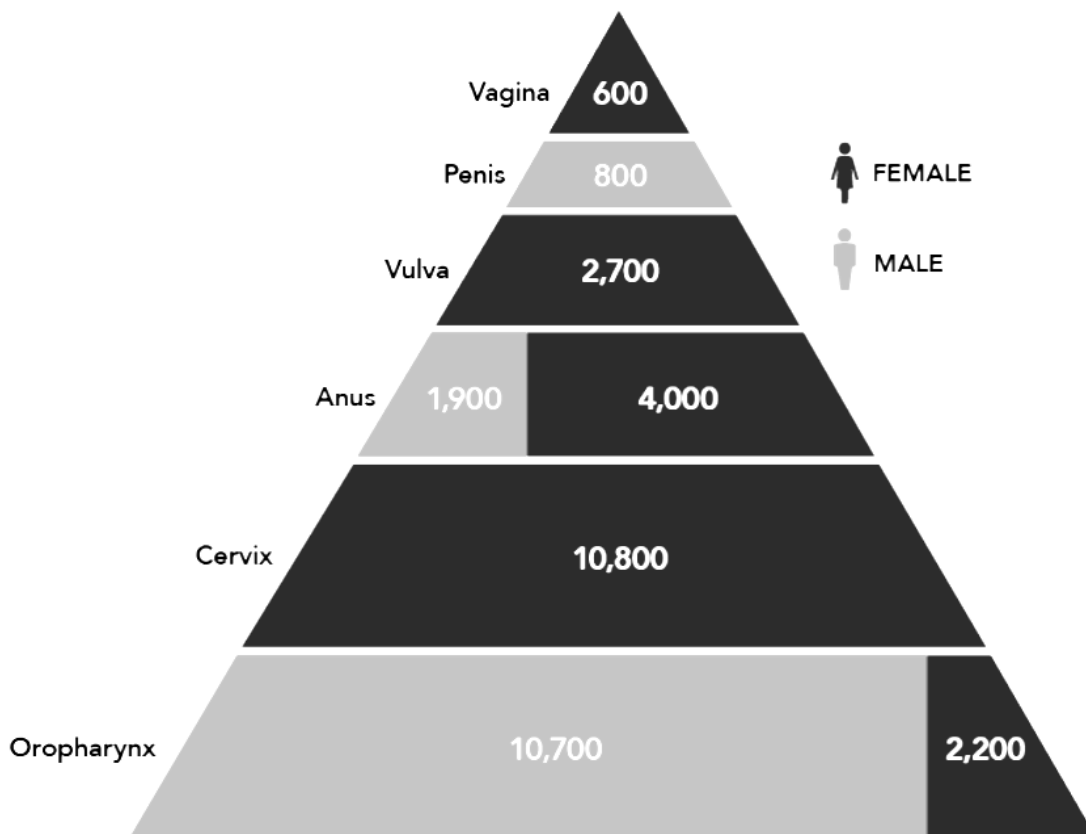


Fig. 1.8

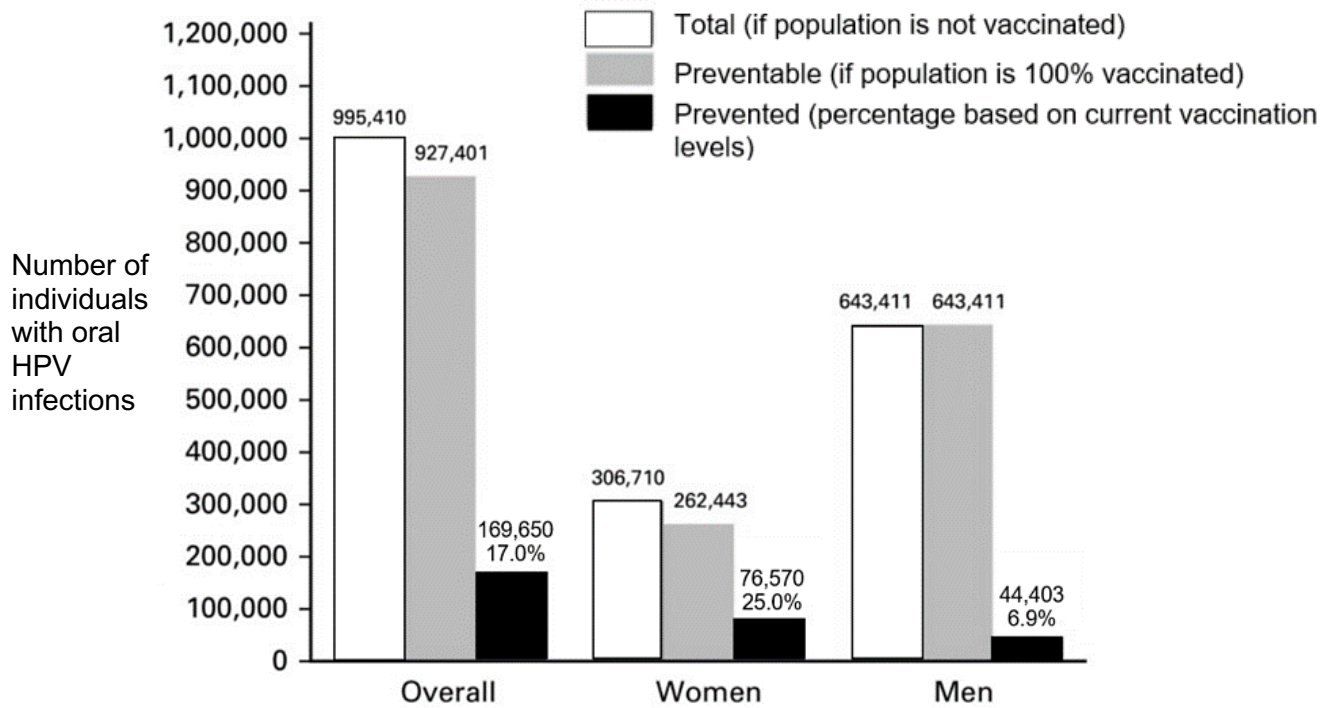


Fig. 1.9

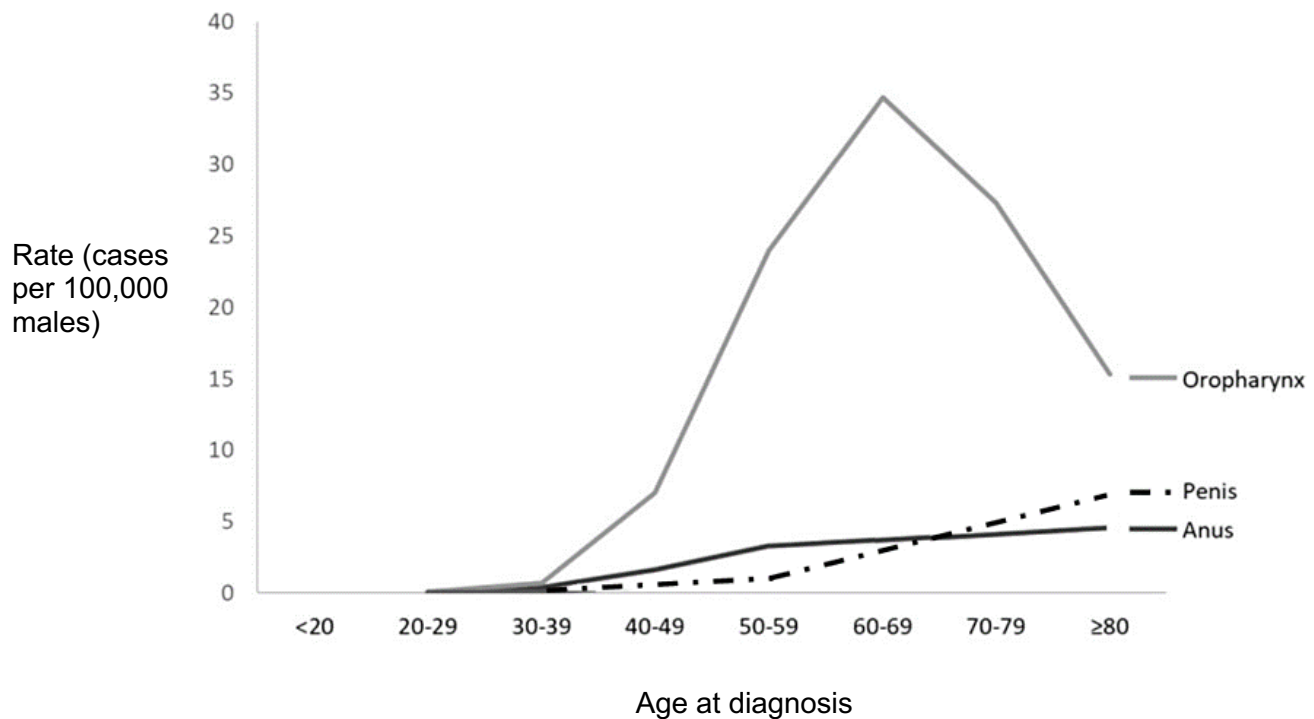


Fig. 1.10

[Turn over

- (f) With reference to Fig. 1.7, Fig. 1.8, Fig. 1.9 and Fig. 1.10, discuss why HPV vaccination should be encouraged in young males less than 20 years old in the United States population to reduce the risk of HPV-associated oropharyngeal cancer.

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[Total: 30]

- 2** The venom of a snake contains many protein toxins which can damage the tissues of a victim who has been bitten. The snake bite can lead to significant disability or death within hours, and an antivenom would be necessary for treatment. The following steps describe how an antivenom is traditionally produced.

- The venom of a snake is collected.
- An animal, often a horse, is injected with a controlled quantity of the venom.
- The horse's blood is withdrawn and the antibodies produced in response to the protein toxins are isolated.
- The isolated antibodies are purified and formulated as an injection.

The antivenom produced is effective only against the species of snake from which the venom is obtained.

- (a) (i)** State the type of immunity that is conferred by the antivenom when administered to the snake-bite victim.

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- (ii)** Describe how antibodies in the antivenom may reduce the harmful effects of toxins in the snake venom.

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- (iii)** Explain why a particular antivenom is effective only against a specific species of snake.

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[Turn over

- (iv) Suggest why injection with vaccines containing inactivated protein toxins is not an effective treatment for a snake-bite victim.

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Fig. 2.1 shows how the blood withdrawn from the horse is centrifuged to obtain three distinct layers – the blood plasma, a layer of white blood cells, and a layer of red blood cells.

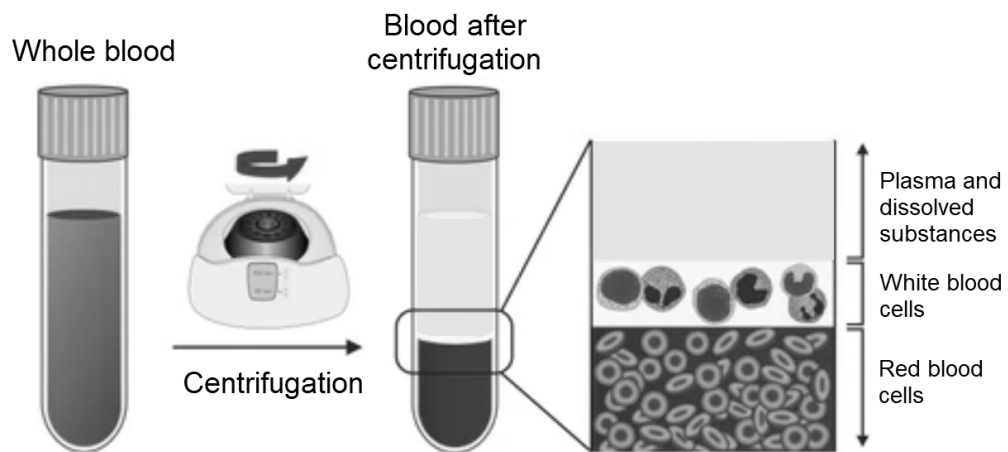


Fig. 2.1

- (b) State the layer which the antivenom antibodies would be found in and explain your answer.

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Due to a global shortage in supply of antivenoms, scientists are exploring the use of genetically modified *Escherichia coli* to mass produce antivenom antibodies. The genes coding for the antibodies can be inserted into *E. coli* by transformation.

- (c) Suggest one advantage and one challenge of using *E. coli* for antivenom production over the traditional method.

Advantage

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Challenge

..... [2]

[Total: 10]

[Turn over

- 3 Malaria is a mosquito-borne disease caused by a unicellular parasite, *Plasmodium*, which spends a part of its life cycle in a mosquito and a part of it in a human. The mosquito transmits the *Plasmodium* to a human when it feeds on human blood. In the low-lying coastal country of Belize, where malaria is a serious problem, studies have been made to determine the environmental factors which affect the incidence of the disease.

156 villages were studied over a ten-year period. Fig. 3.1 highlights the incidence of malaria in the different districts of Belize.

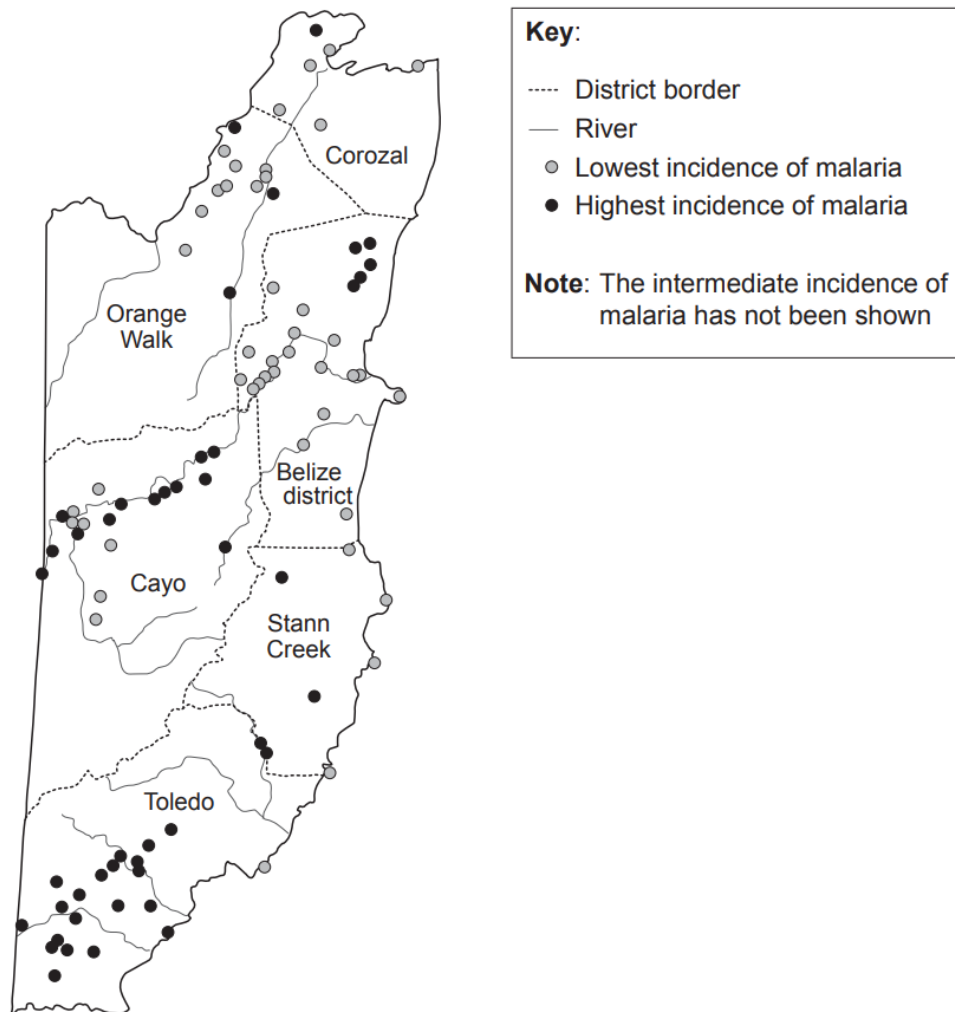


Fig. 3.1

- (a) (i) Based on the studies conducted, explain why the association of rivers with a high incidence of malaria is inconclusive.

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- (ii) Suggest possible factors why the incidence of malaria is low in Corozal compared to other districts.

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Table 3.1 shows the total surface area for different types of land use in the country of Belize.

Table 3.1

Land Use		Thousands of hectare
Agricultural	Arable land (suitable for growing crops)	70
	Permanent crops	32
	Pastures	50
Non-agricultural	Forest area	1412
	Other lands	717

- (b) (i) Calculate the percentage of land used for agricultural purposes in Belize.

Percentage of land used for agriculture = [2]

[Turn over

Climate change is affecting agricultural production and productivity in Belize. Mean annual temperatures have increased at an average rate of 0.1°C per decade since 1960 and climate projections suggest that temperatures could rise by another 1.8°C by 2050. Rainfall is likely to fall throughout the country, with decreases ranging from 7% in the northern zone to 10% in the southern zone. It has also been reported that some farmlands in the southern zone have a high incidence of malaria and pests.

- (ii) Explain why converting some of the forests to farmland will worsen the effects of climate change.

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- (iii) Suggest changes Belize can adopt in its agricultural practices to improve agricultural productivity amidst climate change.

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[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on the separate writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

- 4 (a)** Explain how the structure of proteins that bind to mRNA is linked to their roles in the regulation of eukaryotic gene expression at post-transcriptional and translational levels. [15]
- (b)** Explain the significance of bond formation in the process of DNA replication and describe how the end-replication problem arises in eukaryotes. [10]

[Total: 25]

- 5 (a)** Describe how eukaryotic ribosomes are formed and explain the role of ribosomes in translation. [15]
- (b)** It is known that the first eukaryotic cells were unicellular and had evolved from a prokaryotic ancestor.

Explain the functions of telomeres and suggest how telomeres may have originated and evolved in early eukaryotes. [10]

[Total: 25]